

TECHNICAL ASSESSMENT

Clinical Named Entity Recognition (NER) System: Design Brief

For the role of NLP Engineer

ISSUED BY	MedVerse AI, Clinical Language Systems Group
ROLE	NLP Engineer (Health & Clinical Text)
ASSESSMENT TYPE	Take-home design and implementation exercise
DOMAIN	Healthcare — diseases, medications & symptoms
DATE OF ISSUE	2 July 2026
TIME ALLOWED	72 hours from receipt

1. CONTEXT

This assessment is for an NLP Engineer role at MedVerse AI. The team builds systems that read and understand clinical and health-related text — including tools that automatically identify and classify medical entities inside unstructured notes, discharge summaries, and patient messages. The task below is representative of the work the team does day to day. It is used to check how well a candidate can design, build, and explain a working solution over free-form healthcare text.

2. PROBLEM STATEMENT

Design and implement a system that automatically identifies and classifies named entities from healthcare text. The system shall recognise clinically meaningful entities such as **PERSON** (patient or clinician), **AGE**, **DISEASE / CONDITION**, **SYMPTOM**, **MEDICATION**, **DOSAGE**, **FREQUENCY**, **ROUTE** (of administration), **PROCEDURE**, **LAB_TEST / RESULT**, **ANATOMY** (body part or organ), **MEDICAL_DEVICE**, and **DATE / TIME**.

3. OBJECTIVE

Given an input passage of clinical or health-related text, the system shall identify all named entities present and assign each one the correct entity label. The extracted entities shall be accurate and shall be clearly presented alongside the original text. Where the text expresses that a finding is absent (for example, “no chest pain” or “denies fever”), the system shall not report the negated finding as a present condition or symptom.

4. FUNCTIONAL REQUIREMENTS

4.1 Text Preprocessing

The system shall clean and preprocess the input text before entity detection, including case normalisation where appropriate, removal of unwanted symbols and formatting artefacts, sentence segmentation, and tokenisation. Preprocessing shall preserve clinically important characters (for example, dosage units such as “mg”, “ml”, and “%”) rather than stripping them.

4.2 Named Entity Detection

The system shall detect entities in the text using an NLP model, such as spaCy, a fine-tuned BERT variant (for example, a biomedical or clinical BERT), or a Hugging Face Transformers token-classification model. Candidates may fine-tune a pre-trained model on a public clinical or biomedical dataset.

4.3 Entity Classification

The system shall classify each detected entity into one of the predefined categories set out in the problem statement. Common clinical abbreviations and shorthand (for example, “BP”, “HTN”, “q8h”) shall be handled as far as is reasonable, and the chosen approach shall be explained.

4.4 Entity Extraction Output

The system shall return each detected entity together with its entity text, entity label, start position, and end position (character offsets) in the original text.

4.5 Negation & Context Handling

The system shall distinguish present findings from negated or absent ones. A negated condition or symptom shall be flagged as such rather than reported as a positive finding.

4.6 Visualization

The system shall highlight entities in the original text, using a different colour for each entity category, with a clear legend mapping colours to labels.

4.7 Generalisation

The system shall not be tuned to a single note or dataset and shall work on unseen healthcare text from different sources (for example, clinician notes, discharge summaries, and patient-written messages).

4.8 User Interaction

The user shall be able to enter text and receive entity predictions in response, through a simple interface (a command-line tool, notebook, or lightweight web app is acceptable).

5. INPUT AND OUTPUT SPECIFICATION

Input: Plain healthcare text entered by the user (for example, a short clinical note or a patient message).

Output: The detected entities, with labels, character positions, and a present/negated flag, together with the original text shown with entities highlighted.

Illustrative example. For the input “Mr. Rao, 58, was started on Metformin 500 mg twice daily for type 2 diabetes; reports mild nausea but denies chest pain.” a correct system would tag **Mr. Rao** (PERSON), **58** (AGE), **Metformin** (MEDICATION), **500 mg** (DOSAGE), **twice daily** (FREQUENCY), **type 2 diabetes** (DISEASE / CONDITION, present), **nausea** (SYMPTOM, present), and **chest pain** (SYMPTOM, negated).

6. EVALUATION CRITERIA

REF	CRITERION	DESCRIPTION
6.1	Entity Detection Accuracy	The system correctly identifies the medical entities present in the input text, with minimal omissions.
6.2	Entity Classification Accuracy	Detected entities are assigned to the correct category, such as MEDICATION, DISEASE / CONDITION, or DOSAGE.
6.3	Entity-Level Evaluation	Performance is reported using entity-level precision, recall, and F1 on a held-out sample, with the method clearly described.
6.4	Negation Handling	Absent or denied findings are correctly separated from present findings rather than reported as positive.
6.5	Robustness	The system handles messy, informal, or noisy clinical text (abbreviations, typos, inconsistent casing) without failing or returning incorrect spans.
6.6	Generalisation	The system performs reliably on healthcare text from sources not used during development.
6.7	Code Quality	The code is clear, organised, reproducible, and easy to run.

7. SUBMISSION REQUIREMENTS

Candidates shall submit the following:

- Source code
- Report
- README
- Output screenshots
- Short explanation video

The explanation video is required. It shall cover: (a) the overall system architecture; (b) the data preprocessing steps; (c) how entities are extracted, classified, and checked for negation; and (d) a complete walkthrough of one example, from input text to final prediction. A submission without the video will be treated as incomplete.

8. TIMELINE

The assessment shall be completed and submitted within 72 hours from the time of issue. Late submissions will not be considered unless an extension is agreed in writing in advance.

Note to candidate: You may use any public healthcare or biomedical text dataset (for example, de-identified clinical note corpora or biomedical NER datasets) and any common Python libraries (for example spaCy, scispacy, Hugging Face Transformers, Datasets, PyTorch, and NLTK), as long as you list them in your explanation. Do not use any real, identifiable patient data; use only de-identified or synthetic text. This exercise is intended as an engineering demonstration and is not a medical device; outputs must not be presented as clinical advice. If anything is unclear, please ask before you start.